

Practice Quiz: Learning from Failure and Success

Part A: Multiple Choice

1. Parameter learning differs from structure learning in that: A) Parameter learning changes model values; structure learning changes the model's causal architecture B) Parameter learning is harder than structure learning C) Parameter learning applies only to software; structure learning applies only to hardware D) Parameter learning is always sufficient; structure learning is never needed
2. The first step in a post-mortem analysis should be: A) Assigning blame to team members B) Establishing the raw facts of what happened, without interpretation C) Determining the financial impact of the failure D) Deciding whether to continue the project
3. A failure mode catalog is most valuable because: A) It satisfies regulatory requirements B) It prevents repeated mistakes and enables anticipation of failure modes in new designs C) It is required for patent applications D) It eliminates the need for future testing
4. Learning from success is often neglected because: A) Successes do not contain useful information B) Inventors celebrate and move on without analyzing why the outcome was positive C) Success analysis is technically impossible D) Failures are always more interesting than successes
5. The abstraction ladder helps with transfer learning by: A) Making all lessons more vague B) Generalizing specific observations into principles applicable across different contexts C) Eliminating the need for specific observations D) Replacing expert judgment with rules
6. Toyota's andon cord system treats every quality problem as: A) A reason to fire the responsible worker B) A learning event that triggers immediate investigation, root cause analysis, and documented countermeasures C) A minor inconvenience to be logged later D) A signal to increase production speed

7. The WD-40 case illustrates that: A) Most inventions succeed on the first attempt B) Systematic documentation and learning from each failed formulation enabled convergence on the correct solution C) Failure is always random D) Chemical engineering does not require testing

Part B: Short Answer

1. A prototype robot arm works perfectly in 9 out of 10 tests but fails intermittently in the 10th. Describe how you would conduct a post-mortem on the intermittent failure. What makes intermittent failures particularly challenging for learning? How does the distinction between parameter and structure errors apply?
2. You have successfully completed a project — your invention works and is ready for production. Describe a success analysis process you would conduct to extract maximum learning. What are the risks of skipping this analysis?
3. Using the abstraction ladder, take the lesson "my 3D-printed hinge broke after 50 cycles of use" and abstract it through five levels from specific observation to meta-principle. Explain at which level the lesson becomes most useful for (a) fixing this specific prototype and (b) improving all future projects.